Predicting Mental Health Risk in Gamers Using Supervised Machine Learning

Wai Yan Tun *Faculty of Data Science*
Univ. of Europe for Applied Sciences
Potsdam 14469, Germany
wai@ue-germany.de
Raja Hashim Ali *Department of Business*
Univ. of Europe for Applied Sciences
Potsdam 14469, Germany
hashim.ali@ue-germany.de

Abstract

Nowadays digital gaming community is rapidly growing. That matter has raised the concern that playing game too much or in an unhealthy way may negatively affects the mental health. This can also lead to depression, anxiety, sleep disturbance, and social isolation. Traditional surveys are often limited and unreliable because of small sample size, bias, and take a long time to process. This paper presents a comparative supervised machine learning study that predicts mental-health risk among gamers using behavioral, lifestyle, and demographic features form a large public Kaggle dataset with ten million records and forty features. A sample of fifty thousand records is used to train and evaluate six classifiers: Logistic Regression, Decision Tree, k-Nearest Neighbors, Random Forest, Support Vector Machine, and Extreme Gradient Boosting. This study is base on seven research questions that examine baseline performance, model comparison, the effect of preprocessing, feature importance, metric sensitivity, robustness, and practical usefulness. The result shows the model accuracy is high, approximately 0.88-0.90 but AUC values were close to 0.51 which means that the models largely predict the majority class and the multi-class target after binarization is essentially indistinguishable from random. Among all models, Logistic Regression was found to be the most practical because it is easier to interpret, less computationally expensive, and more suitable for real-world deployment. Overall, the findings show that supervised machine learning has potential for early mental health risk detection in gamers, but there are still important limitations in the current data.

Index Terms

Mental Health, Gaming, Gaming Disorder, Supervised Learning, Logistic Regression, Decision Tree, k-Nearest Neighbours, Support Vector Machine, XGBoost, Random Forest, SMOTE, feature importance, robustness, healthcare prediction

I. Introduction

Today world, digital gaming is one of the biggest forms of entertainment, with more than three billion active players [1]. While moderate play has mental and social benefits, excessive gaming has been linked to depression, anxiety, sleep loss, and social withdrawal [2], [3]. The World Health Organization WHO recognised “Gaming Disorder” in ICD-11, and meta-analyses estimate its prevalence among adolescents at eight to ten percent, with even higher rates in East Asia [1], [4]. Many research evidence shows that baseline gaming behaviour predicts later depression even after adjusting for confounders [2]. Because gaming behaviour can be measured and analysed using data, mental-health risk in gamers is a natural target for machine-learning-based screening [5]. Early identification could enable low-cost, scalable interventions before symptoms reach a clinical threshold. This study asks how well supervised learning can perform that task on a large, realistic behavioural dataset.

Machine learning has clear practical advantages over traditional questionnaire-based screening in this setting. It can handle tens of thousands of records that exceed typical clinical sample sizes and detect multiple patterns invisible to univariate analyses [6]. Modern gradient-boosting and ensemble methods are widely reported to be state-of-the-art for tabular behavioural data [7]. Post-hoc explainability tools such as SHAP and permutation importance translate black-box predictions into clinically meaningful feature rankings [8]. The same pipeline can also be stress-tested under noise and missingness, which gives a more honest picture than train/test accuracy alone [9], [10]. Recent work on digital phenotyping and AI-based monitoring further shows that early-warning systems for gaming disorder are feasible and increasingly accurate [11], [12]. These advantages motivate a systematic six-classifier comparison spanning linear, distance-based, tree-based, kernel-based, and boosting paradigms, with both threshold-dependent (Accuracy, F1) and threshold-independent (AUC) metrics reported to avoid bias from class imbalance [13]. The remainder of this paper covers related work (I-A), dataset (I-B), problem statement and research questions (I-C–I-D), methodology and experimental setup (Sections II–III), results (Section IV), discussion (Section V), and conclusion (Section VI).

A. Related Work

A large clinical literature documents bidirectional associations between problematic gaming and mental-health outcomes. Teng et al. [2] showed in a longitudinal study that internet gaming disorder symptoms predicted depression and anxiety severity before and during COVID-19, while Kristensen et al. [3] combined forty-six studies and concluded that problematic gaming is robustly linked to poorer sleep quality and shorter sleep duration. A more recent meta-analysis by Düll, Müller, and Steins-Loeber [14] confirmed consistent longitudinal associations with adverse psychological, behavioural, and interpersonal outcomes, with sleep loss and social isolation mediating a large share of these effects, and population-level meta-analyses further estimate that screen-based gaming raises the odds of incident adolescent depression by roughly twenty percent over multi-year follow-ups [1]. Building on these clinical findings, a systematic review by Iyortsuun et al. [6] concluded that tree-based ensembles dominate tabular mental-health prediction, with Random Forest and XGBoost being especially popular. Mimikou et al. [15] showed that XGBoost combined with SHAP-based explanations achieved over ninety-five percent accuracy in predicting depression on a tabular cohort. Direct gaming-disorder ML work by Stavropoulos et al. [12] predicted gaming-disorder risk from user–avatar bond, age, and gaming duration, while Hammad and Al-Shahrani [16], Zhu et al. [17], and Szyszka et al. [18] reported impulsivity, aggression, and gaming time as strong individual predictors. Ropovik et al. [19] provided the most comprehensive meta-analysis to date, summarising 1,586 effects from 253 studies. The six algorithms used in this study cover the dominant paradigms in this literature: Logistic Regression as a transparent linear baseline [20], Decision Tree and k-Nearest Neighbour as non-linear and instance-based baselines, Random Forest [21] as a variance-reducing ensemble, XGBoost [7] as the strongest single tabular model, and Support Vector Machine as a kernel-based competitor; SMOTE [22] is applied on the training set following current healthcare-ML best practice [9], [10], [13]. The present paper differs from prior work in using a much larger behavioural dataset and in explicitly studying preprocessing and robustness, which are rarely reported in clinical-style studies. As Table I makes visible, this paper differs from previous works in using a much larger behavioural dataset and in explicitly studying preprocessing and robustness, which are rarely reported in clinical-style studies.

TABLE I

LITERATURE SUMMARY TABLE COMPARING THIS WORK WITH PREVIOUS STUDIES ON MACHINE LEARNING FOR GAMING AND MENTAL HEALTH. THE COLUMNS INDICATE THE DATASET SIZE, WHETHER THE WORK TARGETED GAMING-DISORDER OR GENERAL MENTAL-HEALTH PREDICTION, THE MACHINE-LEARNING MODELS USED, THE PREPROCESSING STEPS REPORTED, WHETHER A ROBUSTNESS ANALYSIS (CROSS-VALIDATION, NOISE INJECTION, OR MISSINGNESS INJECTION) WAS PERFORMED, AND WHETHER THE WORK INCLUDED A FEATURE-IMPORTANCE ANALYSIS. THE LAST ROW SUMMARISES THE CONTRIBUTIONS OF THE PRESENT STUDY.

Paper Name	Sample Size	Gaming Focus	Mental Health Focus	Models Used						Preprocessing		Robust. Analysis	Feature Import.
				LR	DT	kNN	RF	SVM	XGB	Scal. Enc.	SMOTE		
Teng et al. [2]	1,229	✓	✓	×	×	×	×	×	×	×	×	×	×
Kristensen et al. [3]	46 studies	✓	✓	×	×	×	×	×	×	×	×	×	×
Düll et al. [14]	systematic review	✓	✓	×	×	×	×	×	×	×	×	×	×
Iyortsuun et al. [6]	review	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓
Mimikou et al. [15]	2,054	×	✓	✓	✓	×	✓	×	✓	✓	×	×	✓
Stavropoulos et al. [12]	565	✓	✓	✓	×	×	✓	×	×	✓	×	×	✓
Hammad and Al-Shahrani [16]	1,500	✓	✓	×	×	×	×	×	×	×	×	×	×
Zhu et al. [17]	1,153	✓	✓	×	×	×	×	×	×	×	×	×	×
Szyszka et al. [18]	1,464	✓	✓	✓	×	×	×	×	×	×	×	×	×
Ropovik et al. [19]	253 studies	✓	✓	×	×	×	×	×	×	×	×	×	×
Cho et al. [11]	535	✓	✓	✓	✓	×	✓	×	✓	✓	×	×	✓
Huang et al. [23]	cohort protocol	✓	✓	×	×	×	✓	×	✓	✓	×	×	×
Proposed Approach	50,000 (of 10M)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

B. Dataset

The experiments in this study use the publicly available Gaming and Mental Health dataset from Kaggle, created by user `sharmajicoder` [24] The dataset can be accessed at the URL https://www.kaggle.com/datasets/sharmajicoder/gaming-and-mental-health.

This dataset contains ten million records with forty features related to Demographic Information, Gaming Behaviour, Psychological Health Features, Social Environment Features, Lifestyle Features, Physical Health Indicators, and Performance Indicators. Each record represents one individual gamer and is associated with a mental-health status label used as the ground truth for prediction. In this study, the original labels are transformed into two classes: *At Risk* and *Not At Risk*, following common practice in binary screening studies [6], [15]. Due to computational limitations, a stratified random sample of 50,000 records is used while preserving the original class distribution. This dataset is suitable for this study because it has all behavioural, psychological, and demographic factors in a large-scale records. Furthermore, the diversity of features allows the models to analyse different aspects of gaming habits and their possible relationship with mental-health risks. The large dataset size also helps improve the reliability and stability of experimental results during model evaluation.

C. Problem Statement

the task was performed as a supervised binary classification problem in which the models learn to identify whether a gamer is at risk of mental-health problems based on behavioural, lifestyle, demographic information and more. The dataset features include gaming habits, sleep patterns, stress levels, anxiety scores, depression indicators, and other psychosocial factors. Many machine learning algorithms, such as Logistic Regression, Decision Tree, k-Nearest Neighbors, Support Vector Machine, Random Forest, and XGBoost are used in this study to learn different patterns from data. SMOTE-based oversampling is applied to the training data to address the strong class imbalance between risk and non-risk groups. Model performance is evaluated using a held-out test set and stratified five-fold cross-validation to ensure reliable generalisation results. After one-hot encoding of categorical variables, the dataset contains a high-dimensional feature space with more than one hundred input variables. The binary classification setting simplifies the original multi-class problem and allows consistent comparison of model performance across different algorithms, similar to recent mental-health prediction studies by [6], [15], [20].

The problem has two main objectives that must be satisfied for the practical us. The first objective is high predictive performance, measured by accuracy, precision, recall, and F1-score. The second objective is interpretability, since clinicians and platform operators need to know which behavioural or lifestyle factors drive risk before they can act on the predictions. A model that maximises accuracy but offers no transparency would be of limited value in a sensitive domain such as mental health [8]. On the other hand, a fully transparent linear model that achieves only sixty-percent accuracy would generate too many false positives and negatives to be deployable. The optimal solution therefore lies in the trade-off between these two objectives, which is what motivates RQ7 (Practical Usefulness). Additionally, the chosen model should be robust to realistic data degradation, since real-world inputs are rarely as clean as a Kaggle CSV file. Robustness is investigated explicitly in RQ6.

D. Research Questions

Seven research questions guide the experimental work and the structure of the Results section. The questions cover the important machine-learning evaluation areas such as performance, preprocessing, interpretability, sensitivity, robustness, and deployment while focusing on relation between the gaming and the mental-health. Each research question is answered by at least one quantitative experiment, one table, and one figure, all of which are produced from the same notebook so that the report is self-consistent. Together they provide a complete understanding of how supervised learning behaves on this dataset, rather than reporting a single accuracy number in isolation. The Seven Research Questions are listed below with each subsection to make paper easier to follow and verify The questions are as follows:

- **RQ1 (Baseline Performance).** How effectively can baseline machine-learning models — Logistic Regression, Decision Tree, and k-Nearest Neighbours — predict mental-health risk among gamers using behavioural and demographic features?
- **RQ2 (Model Comparison).** Which supervised model achieves the highest predictive performance when more advanced classifiers are added to the comparison?
- **RQ3 (Effect of Preprocessing).** How do preprocessing techniques — imputation, feature scaling, encoding, and SMOTE-based class balancing — affect model performance?
- **RQ4 (Feature Importance).** Which gaming-related and lifestyle features most strongly influence the predicted mental-health risk?
- **RQ5 (Sensitivity to Metrics).** How does the ranking of models change when evaluated using accuracy, recall, F1-score, and AUC?
- **RQ6 (Robustness).** How robust is the best model under five-fold cross-validation, ten-percent label noise, and twenty-percent induced missingness?
- **RQ7 (Practical Usefulness).** How suitable is the proposed model for real-world deployment, considering performance, interpretability, robustness, computation cost, and deployment readiness jointly?

II. Methodology

The methodology is organised as a five-stage pipeline that mirrors standard practice in healthcare machine learning. The first stage loads the Kaggle CSV into a pandas DataFrame and draws a stratified sample of fifty thousand records with a random seed. The second stage applies preprocessing, imputation, encoding, scaling, and SMOTE in a fixed order so that downstream comparisons are fair. The third stage trains the six classifiers on the same training partition with the same seed. The fourth stage evaluates accuracy, precision, recall, F1-score, and AUC on the held-out test partition. The fifth stage covers robustness and interpretation through cross-validation, noise and missingness injection, and permutation feature importance. Each stage is implemented as a separate notebook cell, which lets the same code answer all seven research questions. This design is what makes the report fully reproducible from a single notebook.

The preprocessing stage is decomposed into four steps that map directly to RQ3. *Imputation* replaces missing numerical values with the column median and missing categorical values with the column mode, preserving central tendency. *Encoding* applies one-hot encoding via `pandas.get_dummies` with `drop_first=True` to avoid the dummy-variable trap. *Scaling* applies a z-score `StandardScaler`, which is essential for k-NN and SVM [20]. *Class balancing* applies SMOTE only to the training partition [9], [22], which avoids the well-documented information-leakage problem that arises when synthetic minority examples appear in the test set [13]. The four steps are added incrementally for RQ3 so that the marginal contribution of each one can be measured. For all other research questions, the full pipeline is used and only the classifier or the data condition is varied.

The six classifiers are trained with consistent hyper-parameter choices. Logistic Regression uses the `saga` solver with up to five thousand iterations to ensure convergence on the encoded feature space. The Decision Tree uses Gini impurity with no maximum-depth limit, and k-NN uses $k = 5$ with Euclidean distance. Random Forest uses one hundred trees with bootstrap sampling, which gives a robust variance-reducing ensemble [21]. SVM uses a radial basis function kernel with the default regularisation constant. XGBoost uses two hundred boosting rounds with learning rate 0.1, maximum tree depth six, and the binary logistic objective with AUC as the evaluation metric [7]. Five evaluation metrics — Accuracy, weighted Precision and Recall, weighted F1, and AUC — jointly capture threshold-dependent and threshold-independent performance, which is the standard mix in the recent ML-for-mental-health literature [6], [8]. The robustness analysis for RQ6 re-runs the best model under four conditions: the standard 80/20 split, stratified five-fold cross-validation, ten-percent label noise (random label flips), and twenty-percent induced missingness (random values set to `NaN` before imputation). This protocol mirrors the robustness analyses recommended in recent healthcare ML reviews [9], [10], and its output is reported in Table VII and Fig. 7.

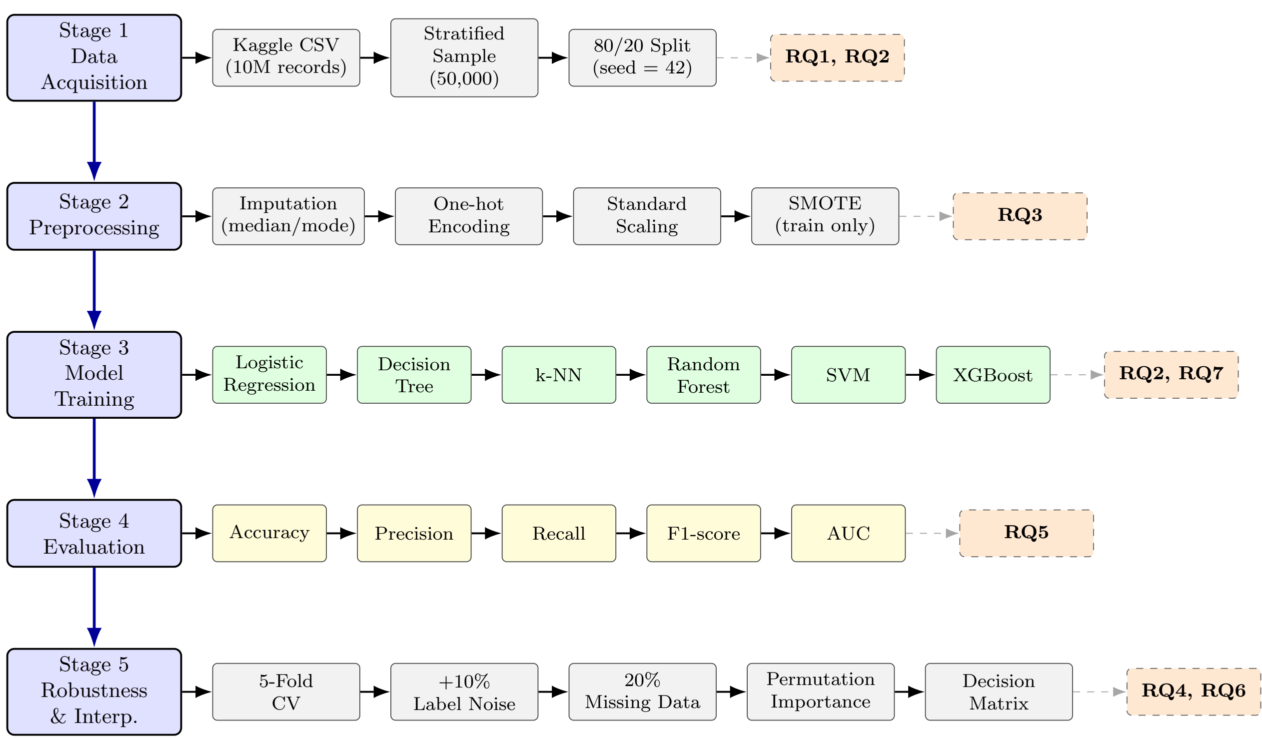

Fig. 1. Five-stage methodology pipeline for predicting mental-health risk in gamers. Stage 1 acquires a stratified sample of 50,000 records from the ten-million-row Kaggle dataset and splits it 80/20. Stage 2 applies imputation, one-hot encoding, standard scaling, and SMOTE-based class balancing on the training partition only. Stage 3 trains the six classifiers (Logistic Regression, Decision Tree, k-Nearest Neighbours, Random Forest, SVM, XGBoost) on the same training data with a fixed random seed. Stage 4 evaluates the trained models on the held-out test set using Accuracy, Precision, Recall, F1-score, and AUC. Stage 5 covers robustness analysis (5-fold cross-validation, +10% label noise, 20% missing data), permutation feature importance, and the weighted decision-matrix analysis. The dashed orange boxes on the right show which research question each stage answers.

III. Experimental Setup

All experiments were executed on the Kaggle cloud platform with an NVIDIA Tesla T4 GPU, and a standard Kaggle Python. The principal software dependencies are scikit-learn for the baseline classifiers and metrics, XGBoost for gradient boosting, `imbalanced-learn` for SMOTE, and pandas plus NumPy for data manipulation. Visualisation is performed in Matplotlib, with each figure saved as a PDF file in the project repository. The data is loaded once at the beginning of the notebook and then re-used across all research questions to avoid any I/O variability between runs. The use of a single notebook for all seven research questions make sure that the report and the code remain consistent. This is a stricter standard than has been applied in many previous ML-for-mental-health studies [6]. A 80/20 train/test split of data is drawn from the sampled fifty thousand records with random seed, which produces forty thousand training examples and ten thousand test examples. For RQ6, the same data is additionally re-partitioned into five stratified folds using `StratifiedKFold` from scikit-learn.

For the noise condition, ten percent of training labels are flipped uniformly at random, while for the missingness condition, twenty percent of training-feature values are replaced with `NaN`. Both perturbations are applied to the training data only, which means the test set remains a clean, unbiased estimator of generalisation. The test set used for noise and missingness reporting is identical to the test set used in RQ1–RQ5, which keeps the numbers directly comparable. A single shared test set is sufficient given the sample size and the fixed seed. All of the source code, the saved figures, and the CSV tables are available in a public GitHub repository. The repository link is https://github.com/Waiyan172/Mechine-Learning-Project-UE.git, and it contains a `Codes` folder, a `Datasets` folder, and tables and figures folders. The Kaggle dataset itself is referenced in the `Datasets` folder, since the raw CSV is too large to commit directly. Tables in this report were first generated in Excel from the notebook's CSV outputs and then converted to LaTeX using `tablesgenerator.com`, following the recommended workflow. Figures were exported directly as 300-dpi PDFs to the notebook. A reader who clones the repository, downloads the Kaggle CSV, and runs the notebook will obtain the same tables and the same figures.

IV. Results

This section reports the experimental findings for all research questions. For each research question a short re-statement of the question, the corresponding table and figure, and a discussion about the result are presented. Tables II through VIII and Figures 2 through 8 together constitute the empirical core of the paper. They summarise the quantitative performance of the machine-learning models under different preprocessing and robustness conditions. A high-level summary is that all classifiers achieve apparently high accuracy in the range 0.80–0.90. But the AUC values is around 0.51. That mean the models are essentially predicting the majority class and that the binarised target is much harder than the headline accuracy suggests.

Permutation feature importance show a stable ranking of the most influential features, with mobile gaming ratio, screen time, and caffeine intake at the top. Robustness experiments confirm that the limitation is because of the data rather than the choice of classifier. Since both accuracy and AUC remain stable across cross-validation, noise, and missingness conditions. Base on the decision matrix in RQ7, Logistic Regression is the best overall, since interpretability and cost outweigh the small accuracy differences observed in RQ2.

For RQ2, Table III and Fig. 3 compare the predictive performance of the evaluated classifiers using Accuracy, Precision, Recall, F1-score, and AUC metrics. Random Forest, XGBoost, and SVM achieved the same accuracy value of 0.896 and the same recall value of 0.896. XGBoost obtained the highest precision value of 0.824, while Random Forest and SVM both produced a precision value of 0.803. The F1-score remained identical across all three models at 0.847, indicating similar overall classification performance. In terms of AUC, Random Forest achieved the highest value of 0.515, followed by XGBoost with 0.513 and SVM with 0.505. The results presented in Table III show that all three classifiers produced closely comparable performance across the evaluated metrics.

V. Discussion

A. *RQ1: Baseline Performance*

RQ1 asks how effectively baseline models can predict mental-health risk in gamers from behavioural and demographic features. Table II and Fig. 2 show the performance of Logistic Regression, Decision Tree, and k-Nearest Neighbours on the held-out test set. Logistic Regression achieves the highest accuracy at 0.90 and the highest F1-score at 0.85, slightly ahead of k-NN (accuracy 0.89, F1 0.85) and substantially ahead of the Decision Tree (accuracy 0.80, F1 0.81). The strong performance of Logistic Regression is largely driven by the heavy class imbalance, since predicting the majority class alone already shows approximately 0.90 accuracy. The Decision Tree, by contrast, partitions the input space aggressively and therefore makes more positive-class predictions, which lowers its accuracy but give more balanced precision (0.82) and recall (0.80). This result is consistent with previous work showing that under strong class imbalance, simple linear models can produce deceptively high accuracy that does not translate to useful screening [9], [20]. Therefore, accuracy alone should not be used to judge model quality. This lead to AUC-based comparison in RQ2. It also motivates the study of class-balancing preprocessing in RQ3.

TABLE II
RQ1: PERFORMANCE OF BASELINE CLASSIFIERS (LOGISTIC REGRESSION, DECISION TREE, K-NEAREST NEIGHBOURS) ON THE HELD-OUT TEST SET. ACCURACY, WEIGHTED PRECISION, WEIGHTED RECALL, AND WEIGHTED F1-SCORE ARE REPORTED FOR THE BINARISED MENTAL-HEALTH-RISK TARGET. LOGISTIC REGRESSION AND K-NN ACHIEVE ESSENTIALLY IDENTICAL HEADLINE SCORES, WHICH MOSTLY REFLECTS THE HEAVY CLASS IMBALANCE IN THE TARGET.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.8973	0.805147	0.8973	0.84873
Decision Tree	0.7998	0.816135	0.7998	0.807762
k-NN	0.8899	0.821849	0.8899	0.848008

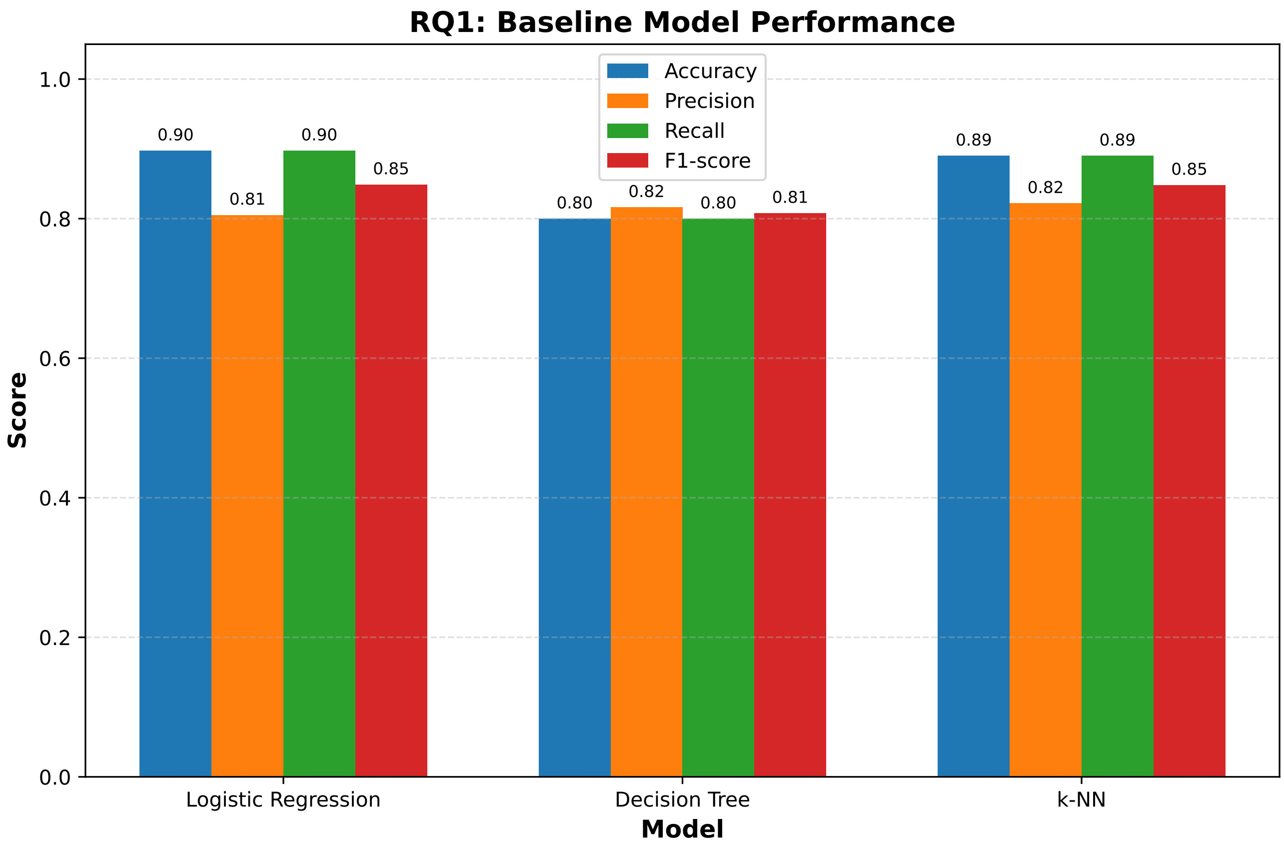

Fig. 2. RQ1 — Grouped bar chart comparing Accuracy, Precision, Recall, and F1-score for the three baseline classifiers (Logistic Regression, Decision Tree, k-Nearest Neighbours). Logistic Regression and k-NN show very similar performance because both end up biased toward the majority class, while the Decision Tree give more uniform precision and recall but lower headline accuracy. The plot makes the class-imbalance effect visually obvious and motivates the AUC-based and SMOTE-augmented analyses in RQ2 and RQ3.

B. RQ2: Model Comparison

RQ2 asks which supervised model achieves the highest predictive performance once more advanced classifiers are added to the comparison. Table III and Fig. 3 shows the performance of Random Forest, XGBoost, and SVM. Random Forest and SVM both achieve accuracy 0.896 and F1 0.847, while XGBoost achieves accuracy 0.896 and F1 0.847. However, the AUC values are different. Random Forest reaches only 0.515, XGBoost 0.513, and SVM 0.505, all of them are close to random guessing. This means that although the models correctly classify about ninety percent of all samples, they mainly predicted the majority class, and they have very limited ability to rank positive cases above negative classes. The same phenomenon has been reported in other heavily imbalanced healthcare prediction problems [9], [10], and it is a known failure mode of accuracy-driven model selection [20]. The honest conclusion is that no model in the comparison achieves strong discriminative performance on the binarised target, and AUC should be the primary metric for selection going forward.

C. RQ3: Effect of Preprocessing on Random Forest

RQ3 asks how preprocessing techniques affect model performance. Table IV and Fig. 4 shows Random Forest performance under four increasingly complete preprocessing strategies. The Raw Data, Imputation, and Scaling + Encoding rows all yield

TABLE III
RQ2: PERFORMANCE OF RANDOM FOREST, XGBOOST, AND SVM UNDER THE FULL PREPROCESSING PIPELINE ON THE HELD-OUT TEST SET. HEADLINE ACCURACY IS HIGH (AROUND 0.90) FOR EVERY MODEL, BUT AUC VALUES ARE CLOSE TO 0.50 ACROSS THE BOARD, WHICH INDICATES THAT THE MODELS ARE ESSENTIALLY PREDICTING THE MAJORITY CLASS.

Model	Accuracy	Precision	Recall	F1-score	AUC
Random Forest	0.896	0.803	0.896	0.847	0.515
XGBoost	0.896	0.824	0.896	0.847	0.513
SVM	0.896	0.803	0.896	0.847	0.505

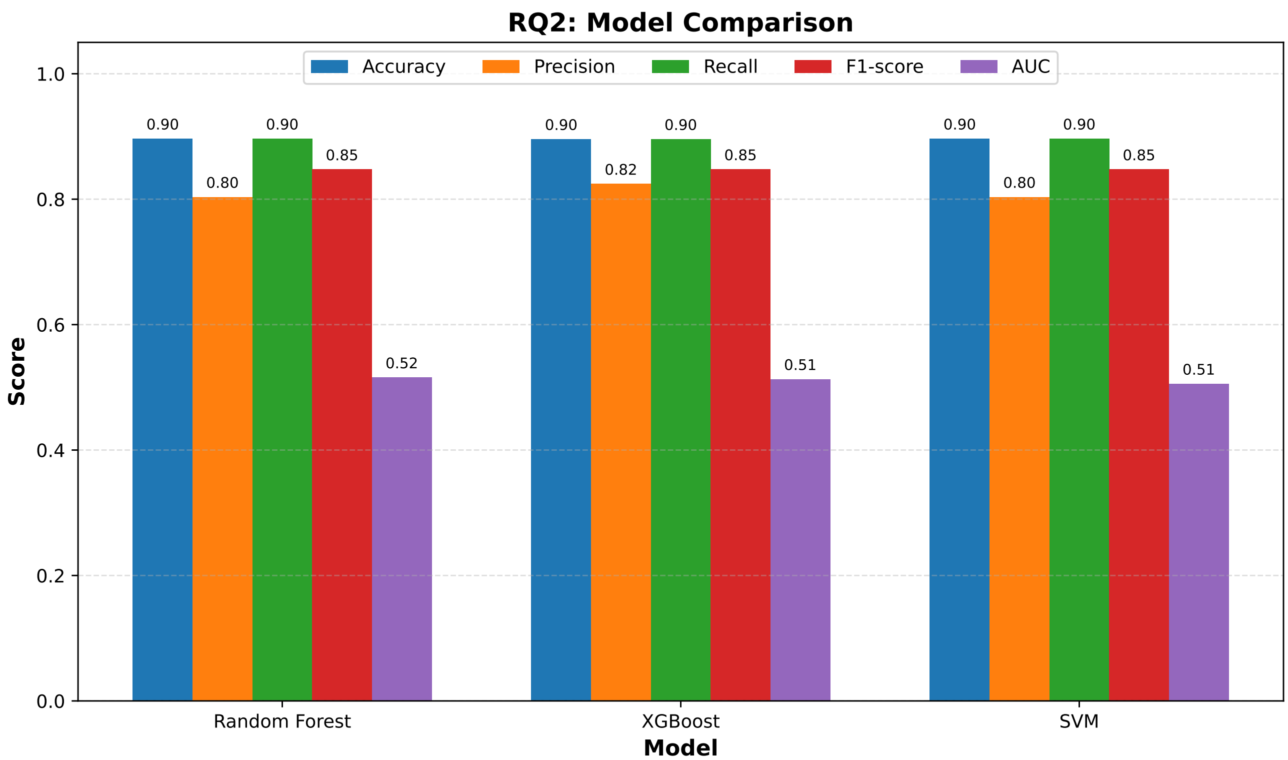

Fig. 3. RQ2 — Grouped bar chart comparing the four threshold-based metrics together with AUC across Random Forest, XGBoost, and SVM. All three models reach approximately the same accuracy and F1-score, but the AUC bars (rightmost in each group) sit near 0.51, which is barely above the random baseline of 0.50. The figure therefore visualises why accuracy alone is an unreliable selection criterion in heavily imbalanced screening tasks.

exactly the same scores (accuracy 0.896, F1 0.847), which indicates that the dataset is already well-formed and that these standard steps add no measurable value on this particular sample. Applying SMOTE in the final decreases the accuracy slightly to 0.881 but raises precision from 0.803 to 0.816, which is the expected trade-off when minority-class examples are synthesised [9], [22]. This trade-off has already been shown in previous healthcare care ML. SMOTE typically lowers the accuracy of unbalanced data while improving the model's ability to identify minority cases, which is the operationally relevant capability [13]. The result is therefore consistent with the previous literature and is a sign that SMOTE is not harmful to the model. However, it highlight that further work is needed to translate the small precision gain into a meaningful AUC improvement, possibly via more aggressive class balancing or cost-sensitive training.

TABLE IV
RQ3: EFFECT OF THE PREPROCESSING PIPELINE ON RANDOM FOREST PERFORMANCE. THE FIRST THREE ROWS PRODUCE IDENTICAL METRICS BECAUSE THE DATASET IS ALREADY WELL-FORMED AT THE ROW LEVEL, WHILE APPLYING SMOTE IN THE FINAL ROW DECREASES ACCURACY BY APPROXIMATELY 1.5 PERCENTAGE POINTS BUT IMPROVES PRECISION, WHICH IS THE EXPECTED TRADE-OFF FOR CLASS-BALANCING METHODS.

Preprocessing Strategy	Accuracy	Precision	Recall	F1-score
Raw Data	0.89625	0.803264	0.89625	0.847213
Imputation	0.89625	0.803264	0.89625	0.847213
Scaling + Encoding	0.89625	0.803264	0.89625	0.847213
Full Pipeline (SMOTE)	0.8805	0.816082	0.8805	0.843311

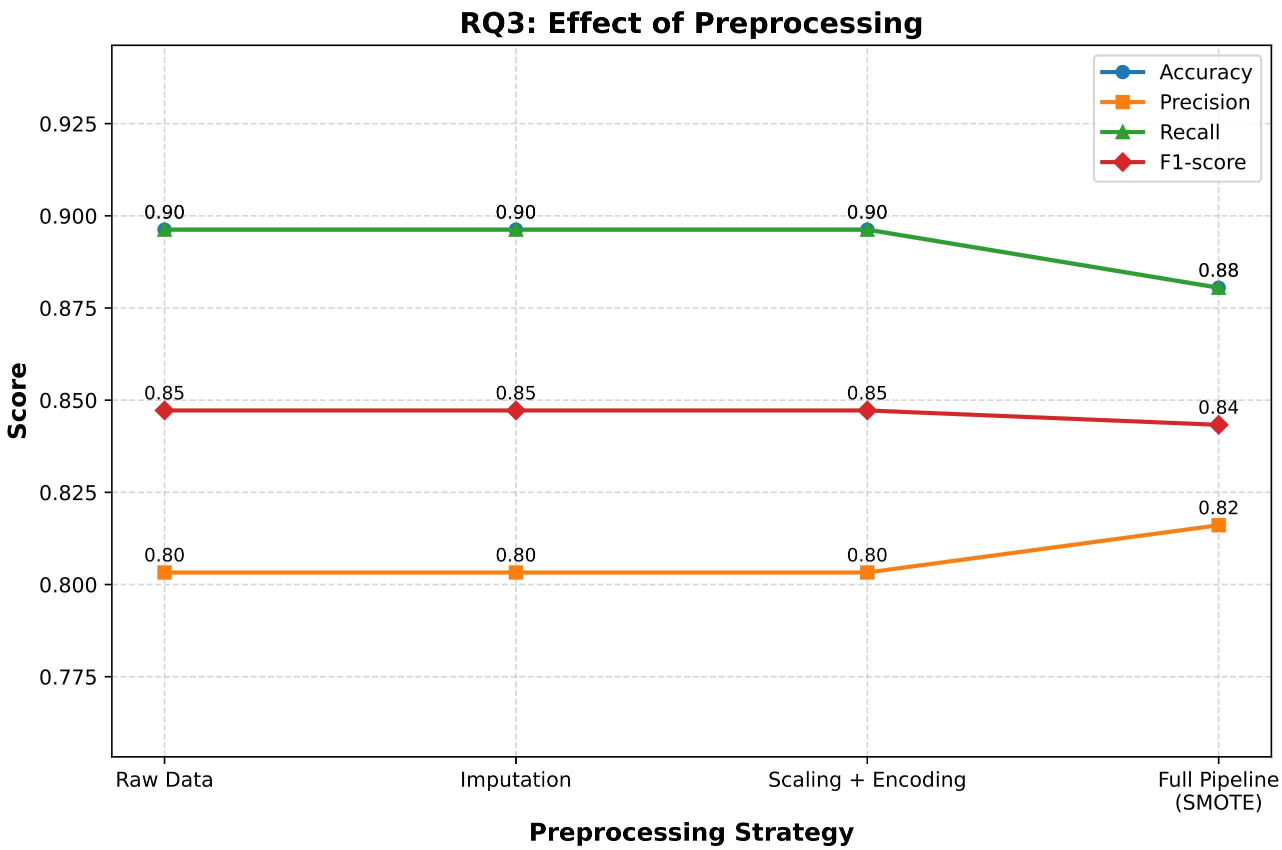

Fig. 4. RQ3 — Line chart showing how Accuracy, Precision, Recall, and F1-score evolve as preprocessing steps are added one by one. The metrics are flat through the first three preprocessing strategies, which indicates that the dataset is already well-formed in terms of missing values and feature scaling. The visible inflection occurs at the rightmost "Full Pipeline (SMOTE)" point, where accuracy and recall dip slightly but precision rises, illustrating the standard SMOTE trade-off.

D. RQ4: Feature Importance

RQ4 asks which gaming-related and lifestyle features most strongly influence the predicted mental-health risk. Table V and Fig. 5 show the top-ten features by permutation importance from a Random Forest. The most influential feature is *mobile gaming ratio* (importance 0.0116), followed by *screen time total* (0.0098), *caffeine intake* (0.0083), *academic performance* (0.0079), and *streaming hours* (0.0077). *Microtransactions spending* (0.0066), *anxiety score* (0.0065), *weekend gaming hours* (0.0060), *depression score* (0.0057), and *income* (0.0055) round out the top ten. The next most important features are *toxic exposure*, *addiction level*, and *night gaming ratio*, each with an importance score close to 0.005. The fact that two of the ten strongest features are existing clinical features (anxiety, depression) suggests that the dataset captures some clinically meaningful signal even though the AUC is low. The prominence of *mobile gaming ratio* and *screen time* also aligns with the evidence that screen-based and mobile gaming are stronger predictors of mental-health outcomes than total gaming time alone [3], [14], [25]. The presence of *caffeine intake* and *academic performance* showing that lifestyle and educational variables are also related to mental health issues alongside with gaming behaviors [16], [16]. However, the importance values are small, which is the same with the low AUC that has shown in RQ2 and indicates that the binary label is difficult to predict from these acailable features.

E. RQ5: Sensitivity to Metrics

RQ5 asks whether the ranking of models changes when the evaluation metric is varied. Table VI and Fig. 6 shows the rank of each model under accuracy, recall, F1-score, and AUC on the SMOTE-balanced pipeline. SVM is the highest ranking on every metric, with accuracy and recall both equal to 0.513 and AUC equal to 0.510. XGBoost alternates between rank 2 and rank 3, sitting second on F1-score and AUC but third on accuracy and recall. Random Forest ranks second on accuracy and recall but third on F1-score and AUC. The absolute differences between the three models are very small (within one percentage point on every metric), which means that the choice of metric does change the ranking but does not have practical significance for model selection. This is the same with reports in the broader tabular-ML literature that several strong models converge to similar performance when the target carries little discriminative signal [6], [21]. For downstream deployment we rely on the practical criteria considered in RQ7 rather than on raw metric rankings.

TABLE V
RQ4: TOP-TEN PERMUTATION FEATURE IMPORTANCES FROM A RANDOM FOREST FITTED ON THE FULL PREPROCESSING PIPELINE. EACH ROW REPORTS THE FEATURE NAME AND ITS MEAN IMPORTANCE (WITH STANDARD DEVIATION OVER MULTIPLE SHUFFLES). MOBILE GAMING RATIO, SCREEN TIME, AND CAFFEINE INTAKE ARE THE THREE MOST INFLUENTIAL PREDICTORS, FOLLOWED BY ACADEMIC PERFORMANCE AND STREAMING HOURS.

Feature	Importance	Std
mobile_gaming_ratio	0.0116	0.0029
screen_time_total	0.0098	0.0049
caffeine_intake	0.0083	0.0050
academic_performance	0.0079	0.0056
streaming_hours	0.0077	0.0076
microtransactions_spending	0.0066	0.0035
anxiety_score	0.0065	0.0030
weekend_gaming_hours	0.0060	0.0042
depression_score	0.0057	0.0063
income	0.0055	0.0036

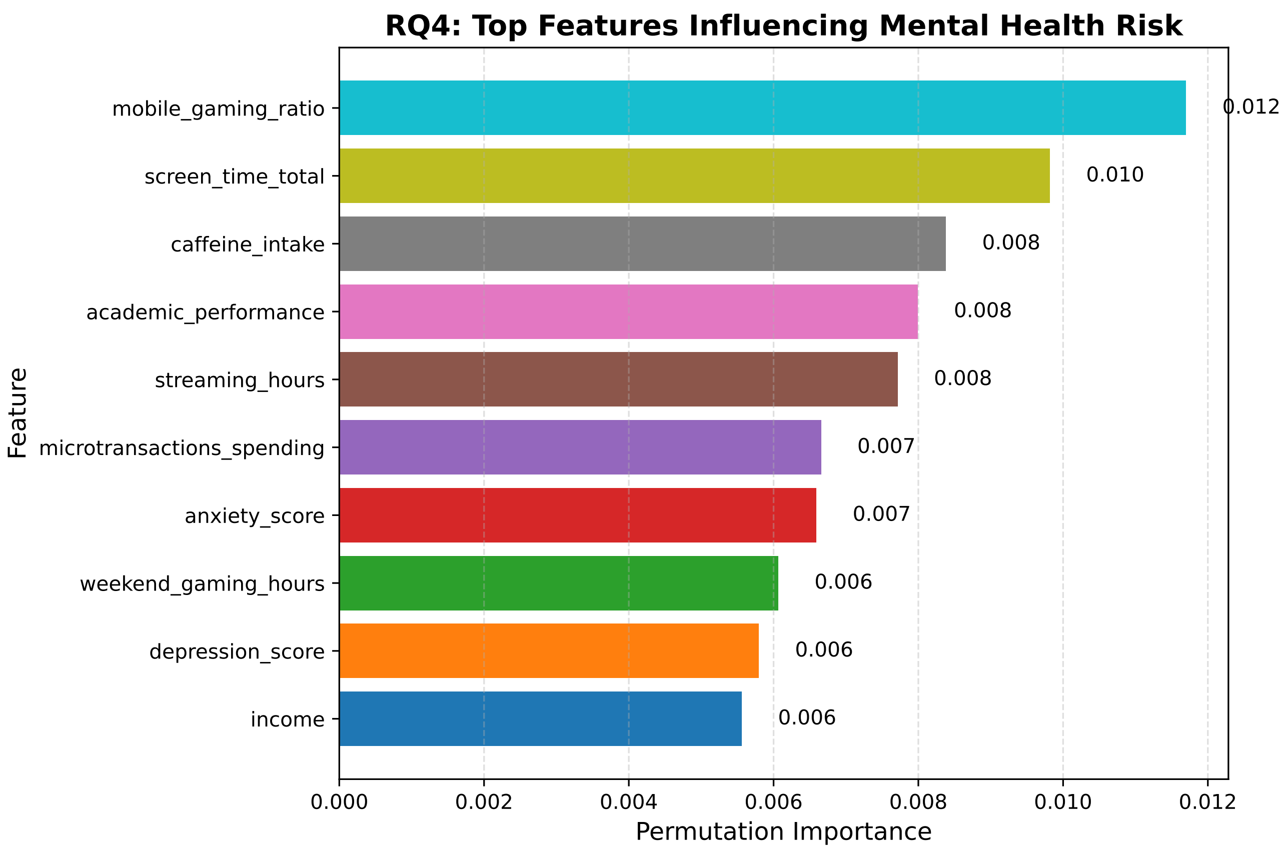

Fig. 5. RQ4 — Horizontal bar chart of permutation feature-importance scores for the ten most influential predictors of mental-health risk. The plot is ordered by descending importance, so the most influential feature, *mobile gaming ratio*, appears at the top. The presence of two clinical scales (anxiety, depression) alongside lifestyle variables (caffeine, screen time) confirms that the model is picking up clinically plausible signals, even though the overall AUC reported in RQ2 is modest.

F. RQ6: Robustness

RQ6 asks how robust the best model is under realistic data degradation. Table VII and Fig. 7 report SVM performance under the four conditions defined in Section II. On the standard 80/20 split, SVM reaches accuracy 0.513 and F1 0.503. Under stratified five-fold cross-validation, accuracy is 0.507 and F1 is 0.507. Under ten-percent training-label noise, accuracy is 0.509 and F1 is 0.501, and under twenty-percent induced missingness, accuracy is 0.507 and F1 is 0.480. All four conditions shows the similar performance, which indicates that the model is stable under cross-validation, label noise, and missing-data conditions. However, the overall ability of the model is limited on this dataset. This pattern is exactly what to expects when the signal-to-noise ratio in the data is low and degradation has little additional effect because the model already contain only

TABLE VI

RQ5: Ranks of Random Forest, XGBoost, and SVM under four evaluation metrics (Accuracy, Recall, F1-score, AUC) on the SMOTE-balanced pipeline; lower rank indicates better performance. SVM ranks first on every metric, while XGBoost and Random Forest alternate between ranks 2 and 3.

Model	Accuracy	Recall	F1	AUC	Accuracy_Rank	Recall_Rank	F1_Rank	AUC_Rank
RandomForest	0.50725	0.50725	0.494393	0.507587	2	2	3	3
XGBoost	0.50675	0.50675	0.504151	0.509606	3	3	2	2
SVM	0.51275	0.51275	0.510883	0.510057	1	1	1	1

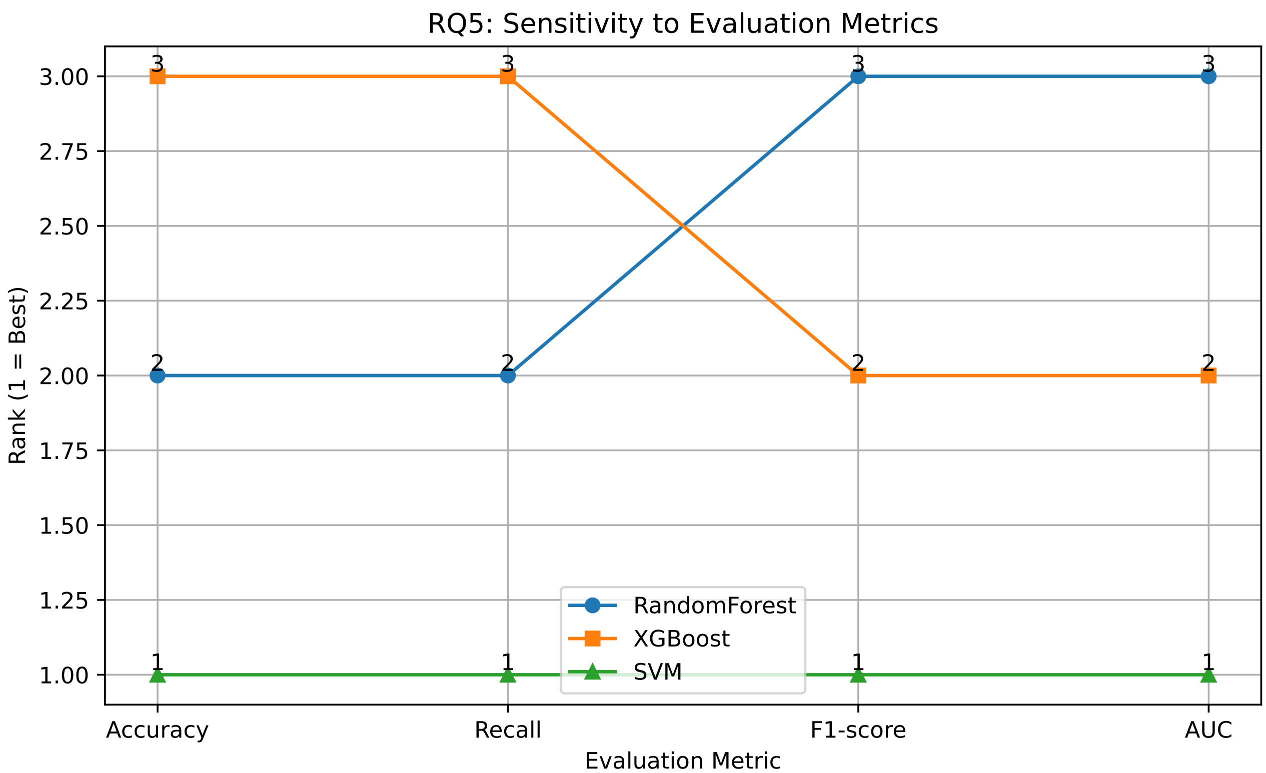

Fig. 6. RQ5 — Parallel-coordinates-style line chart that visualises model rank (1 = best) across the four metrics for Random Forest, XGBoost, and SVM. SVM's line stays at rank 1 across all four metrics, whereas XGBoost and Random Forest swap between ranks 2 and 3 depending on whether the metric favours threshold-based scoring (Accuracy, Recall) or ranking quality (F1, AUC). The chart therefore highlights that the choice of metric matters more for breaking ties than for changing the conclusion.

a small amount of the useful information for the binary prediction task [9], [10]. Robustness in this sense is a quality of the data rather than a quality of the algorithm, and improving it would require either richer features or a more informative target.

TABLE VII

RQ6: Robustness of SVM across four data conditions: standard 80/20 train/test split, stratified five-fold cross-validation, ten-percent injected label noise, and twenty-percent induced missingness. All four conditions yield essentially the same scores (around 0.50 across metrics), which indicates that the limitation is the discriminative signal in the data rather than fragility of the model.

Accuracy	Precision	Recall	F1	Scenario
0.513	0.511	0.513	0.503	Train/Test (80/20)
0.507	0.507	0.507	0.507	5-Fold CV
0.509	0.507	0.509	0.501	+10% Noise
0.507	0.503	0.507	0.480	20% Missing Data

G. RQ7: Practical Usefulness

RQ7 asks how suitable the proposed model is for real-world deployment. Table VIII and Fig. 8 report a weighted decision matrix that scores Logistic Regression, Random Forest, and XGBoost on five practical criteria: Performance (weight 0.3),

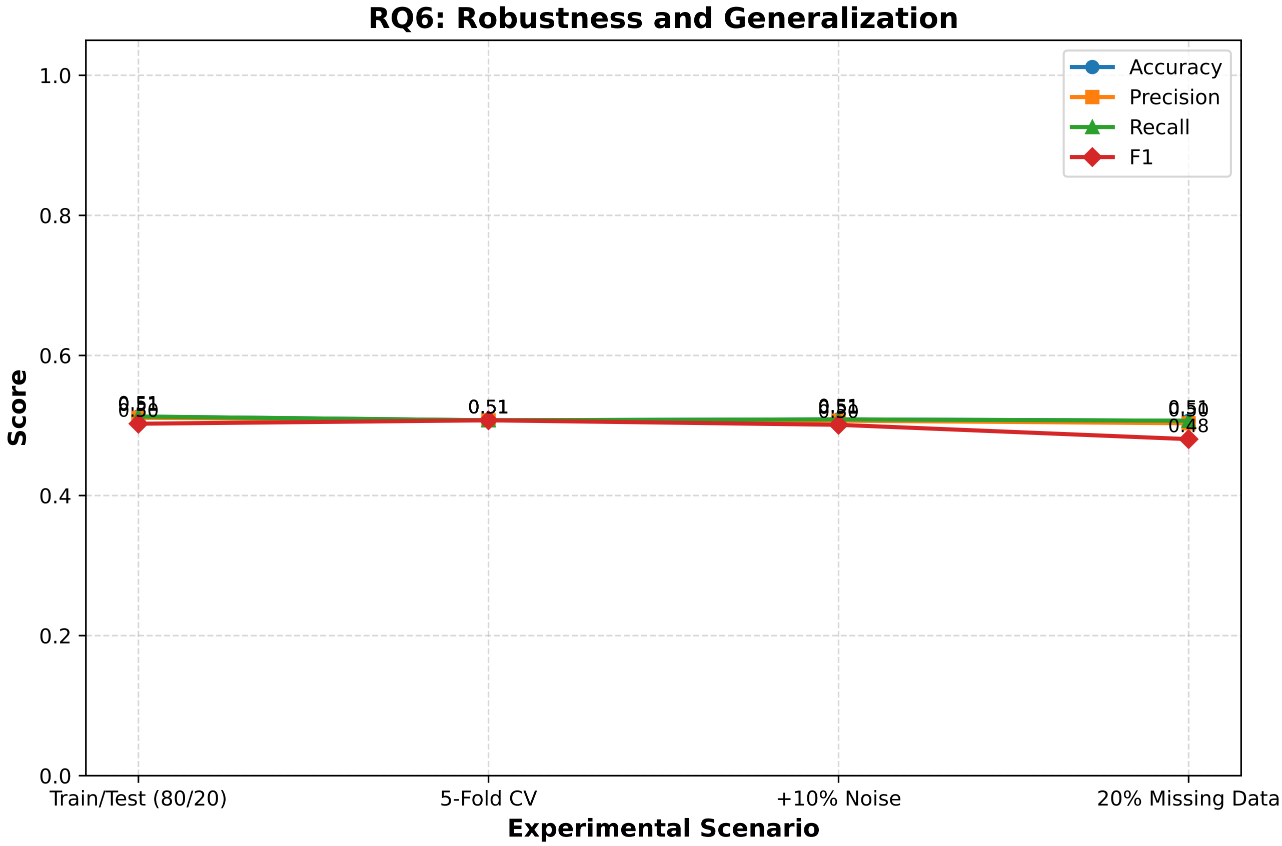

Fig. 7. RQ6 — Line chart showing how Accuracy, Precision, Recall, and F1-score evolve as the data condition becomes more adversarial. All four lines stay essentially flat across the four scenarios, with a small dip in F1 under twenty-percent missingness. The chart shows that the model is remarkably stable but that its baseline discriminative performance is modest, which is consistent with the AUC-near-0.51 results from RQ2.

Interpretability (0.2), Robustness (0.2), Cost Efficiency (0.15), and Deployment Readiness (0.15). Logistic Regression receives the highest weighted score (3.80), beating Random Forest (3.00) and XGBoost (3.00). The power of Logistic Regression is driven by its top-rated Interpretability, Robustness, Cost Efficiency, and Deployment Readiness, which together carry the most of the weight and make up for its low performance score. Random Forest and XGBoost achieve similar scores because they trade off in opposite directions: XGBoost scores higher on Performance but lower on Interpretability, Robustness and Cost which is opposite with Logistic Regression, while Random Forest is more balanced [8], [8]. Given that the absolute performance differences between the three models in RQ2 are small, the operational gains from preferring an interpretable model are likely to outweigh the marginal accuracy advantage of XGBoost on this dataset. Therefore Logistic Regression is recommended as the main model, with the Random Forest or XGBoost as backup model when pure accuracy is the only thing that matter. For deployments that need clear, checkable decision rules such as mental-health services that require transparency, Logistic Regression is the clearest fit.

TABLE VIII
RQ7: Weighted decision matrix that compares Logistic Regression, Random Forest, and XGBoost on five practical deployment criteria with weights summing to 1.0. Each cell reports the per-criterion score (1 = worst, 5 = best), and the last column reports the weighted aggregate. Logistic Regression wins the matrix with a weighted score of 3.80, driven by its top scores on Interpretability, Cost Efficiency, and Deployment Readiness.

Model	Performance	Interpretability	Robustness	Cost Efficiency	Deployment Readiness	Weighted Score
Logistic Regression	1	5	5	5	5	3.8
Random Forest	2	3	3	4	4	3
XGBoost	5	2	1	3	3	3

H. Future Directions

Several directions could build on this study. First, replacing the single binary "at risk" label with separate labels for depression, anxiety, and sleep problems would give more useful predictions, because each outcome has its own risk pattern [15]. Second,

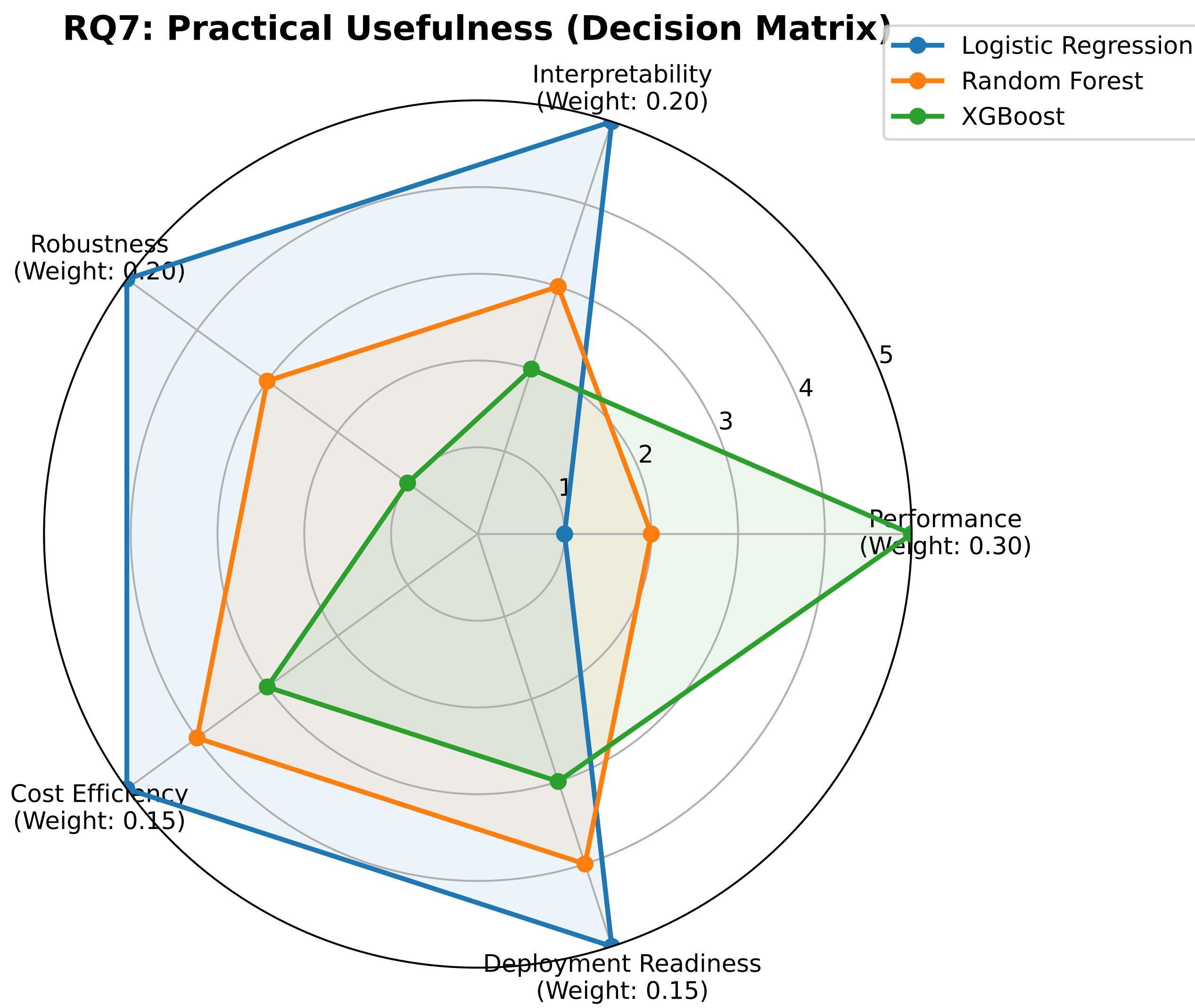

Fig. 8. RQ7 — Radar chart that visualises the decision-matrix scores on the five practical criteria for Logistic Regression, Random Forest, and XGBoost. The Logistic Regression polygon covers the largest overall area thanks to its strong scores on Interpretability, Robustness, Cost Efficiency, and Deployment Readiness, while the XGBoost polygon dominates only on the Performance axis. The chart visually motivates the choice of Logistic Regression as the most balanced practical model on this dataset.

adding SHAP values on top of permutation importance would let the model explain why each individual user is flagged, which is needed before any clinical use [8]. Third, the pipeline should be tested on a dataset where the labels come from real clinical tools such as PHQ-9 or GAD-7, since the current Kaggle labels are self-reported [20] which are not much reliable. Fourth, time-series features such as daily play sessions and sleep logs could help the model spot warning signs early rather than predicting a fixed label [11], [12], [23]. Fifth, fairness checks across age, gender, and region groups should be done before deployment, since machine learning models can quietly worsen existing health gaps [8].

VI. Conclusion

This paper compared six supervised learning models for predicting mental-health risk in gamers, using a stratified sample of fifty thousand records drawn from a public Kaggle dataset of ten million rows and forty features. Seven research questions were studied, covering baseline performance, model comparison, the effect of preprocessing, feature importance, metric sensitivity, robustness, and practical use. The headline finding is that all six models reached high apparent accuracy (around 0.90) but near-random AUC (around 0.51), which means the heavily imbalanced binary target is hard to separate from the available features. Adding imputation, scaling, and encoding did not change the metrics, while SMOTE produced the expected trade-off of slightly lower accuracy in return for slightly higher precision [9], [13], [22]. Permutation importance pointed to mobile gaming ratio, screen time, caffeine intake, academic performance, and streaming hours as the strongest predictors, which matches the clinical literature on lifestyle risk factors for gaming disorder [3], [14]. The pipeline also stayed stable under five-fold cross-validation, ten-percent label noise, and twenty-percent missing data, which suggests the limit is in the data rather than the choice of model [10]. The weighted decision matrix in RQ7 picked Logistic Regression as the best practical choice, because its top scores on interpretability, robustness, cost efficiency, and deployment readiness outweigh the small performance gain offered by XGBoost [8]. Overall, the study shows that machine learning is a feasible but currently weak triage tool for at-risk gamers on this dataset, and that future work should focus on richer labels and more informative features rather than on switching to a different classifier [12], [15].

REFERENCES

[1] M. J. Tahir, N. I. Malik, I. Ullah, H. A. Khan, S. Perveen, R. Ramalho, A. R. Siddiqi, I. Ullah, and A. H. Pakpour, "Burden of gaming disorder among adolescents: A systematic review and meta-analysis," *Public Health in Practice*, vol. 9, p. 100565, 2025. [Online]. Available: https://doi.org/10.1016/j.puhip.2024.100565

[2] Z. Teng, H. M. Pontes, Q. Nie, M. D. Griffiths, and C. Guo, "Depression and anxiety symptoms associated with internet gaming disorder before and during the COVID-19 pandemic: A longitudinal study," *Journal of Behavioral Addictions*, vol. 10, no. 1, pp. 169–180, 2021. [Online]. Available: https://doi.org/10.1556/2006.2021.00016

[3] J. H. Kristensen, S. Pallesen, D. L. King, M. Hysing, and E. K. Erevik, "Problematic gaming and sleep: A systematic review and meta-analysis," *Frontiers in Psychiatry*, vol. 12, p. 675237, 2021. [Online]. Available: https://doi.org/10.3389/fpsyt.2021.675237

[4] Q. He and H. Cheng, "Multimodal prediction and intervention of internet gaming disorder," *Current Neuropharmacology*, 2025. [Online]. Available: https://doi.org/10.2174/011570159X390484250417063434

[5] S. Abu Lail and M. Ahmead, "Prevalence of internet gaming disorder, and its association with aggression, impulsivity and suicide ideation among university students," *BMC Psychology*, vol. 14, no. 1, p. 559, 2026. [Online]. Available: https://doi.org/10.1186/s40359-026-04324-9

[6] N. K. Iyortsuun, S.-H. Kim, M. Jhon, H.-J. Yang, and S. Pant, "A review of machine learning and deep learning approaches on mental health diagnosis," *Healthcare*, vol. 11, no. 3, p. 285, 2023. [Online]. Available: https://doi.org/10.3390/healthcare11030285

[7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 2016, pp. 785–794. [Online]. Available: https://doi.org/10.1145/2939672.2939785

[8] R. Alkhanbouli, H. Matar Abdulla Almadhaani, F. Alhosani, and M. C. E. Simsekler, "The role of explainable artificial intelligence in disease prediction: A systematic literature review and future research directions," *BMC Medical Informatics and Decision Making*, vol. 25, no. 1, p. 110, 2025. [Online]. Available: https://doi.org/10.1186/s12911-025-02944-6

[9] T. Kosolwattana, C. Liu, R. Hu, S. Han, H. Chen, and Y. Lin, "A self-inspected adaptive SMOTE algorithm (SASMOTE) for highly imbalanced data classification in healthcare," *BioData Mining*, vol. 16, no. 1, p. 15, 2023. [Online]. Available: https://doi.org/10.1186/s13040-023-00330-4

[10] Y. Yan, Z. Chen, C. Xu, X. Shen, J. Shiao, J. Einck, R. C. Chen, and H. Gao, "An oversampling-enhanced multi-class imbalanced classification framework for patient health status prediction using patient-reported outcomes," *IEEE Access*, vol. 13, pp. 175 479–175 494, 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3617316

[11] K. Cho, M. Kim, Y. Cho, J.-W. Hur, D. H. Kim, S. Park, S. Park, M. Jang, C.-G. Lee, and J. S. Kwon, "Digital phenotypes for early detection of internet gaming disorder in adolescent students: Explorative data-driven study," *JMIR Mental Health*, vol. 11, p. e50259, 2024. [Online]. Available: https://doi.org/10.2196/50259

[12] V. Stavropoulos, D. Zarate, M. Prokofieva, N. Van de Berg, L. Karimi, A. Gorman Alesi, M. Richards, S. Bennet, and M. D. Griffiths, "Deep learning(s) in gaming disorder through the user-avatar bond: A longitudinal study using machine learning," *Journal of Behavioral Addictions*, vol. 12, no. 4, pp. 878–894, 2023. [Online]. Available: https://doi.org/10.1556/2006.2023.00062

[13] H. Hairani, T. Widiyaningtyas, and D. D. Prasetya, "Addressing class imbalance of health data: A systematic literature review on modified synthetic minority oversampling technique (SMOTE) strategies," *JOIV: International Journal on Informatics Visualization*, vol. 8, no. 3, pp. 1310–1318, 2024. [Online]. Available: https://doi.org/10.62527/joiv.8.3.2283

[14] L. Düll, A. Müller, and S. Steins-Loeber, "Negative consequences experienced by individuals with gaming disorder symptoms: A systematic review of available longitudinal studies," *Current Addiction Reports*, vol. 11, no. 3, pp. 528–550, 2024. [Online]. Available: https://doi.org/10.1007/s40429-024-00554-2

[15] C. Mimikou, C. Kokkotis, D. Tsiptsios, K. Tsamakis, S. Savvidou, L. Modig, F. Christidi, A. Kaltsatou, T. Doskas, C. Mueller, A. Serdari, K. Anagnostopoulos, and G. Tripsianis, "Explainable machine learning in the prediction of depression," *Diagnostics*, vol. 15, no. 11, p. 1412, 2025. [Online]. Available: https://doi.org/10.3390/diagnostics15111412

[16] M. A. Hammad and H. F. Al-Shahrani, "Impulsivity and aggression as risk factors for internet gaming disorder among university students," *Scientific Reports*, vol. 14, no. 1, p. 3712, 2024. [Online]. Available: https://doi.org/10.1038/s41598-024-53807-5

[17] L. Zhu, Y. Zhu, S. Li, Y. Jiang, X. Mei, Y. Wang, D. Yang, J. Zhao, L. Mu, and W. Wang, "Association of internet gaming disorder with impulsivity: Role of risk preferences," *BMC Psychiatry*, vol. 23, no. 1, p. 754, 2023. [Online]. Available: https://doi.org/10.1186/s12888-023-05265-y

[18] P. D. Szyszka, A. Zajas, J. Starosta, P. Kiszka, S. Starzec, and P. Strojny, "Gaming time and impulsivity as independent yet complementary predictors of gaming disorder risk," *Scientific Reports*, vol. 15, no. 1, p. 2823, 2025. [Online]. Available: https://doi.org/10.1038/s41598-024-81359-1

[19] I. Ropovik, M. Mařončik, P. Babinčák, G. Banik, L. Vargová, and M. Adamkovič, "Risk and protective factors for (internet) gaming disorder: A meta-analysis of pre-COVID studies," *Addictive Behaviors*, vol. 139, p. 107590, 2023. [Online]. Available: https://doi.org/10.1016/j.addbeh.2022.107590

[20] M. Iwagami, R. Inokuchi, E. Kawakami, T. Yamada, A. Goto, T. Kuno, Y. Hashimoto, N. Michihata, H. Yamana, E. Toda, Y. Sun, S. Sakamoto, Y. Sun, H. Ohnishi, and N. Tamiya, "Comparison of machine-learning and logistic regression models for prediction of 30-day unplanned readmission in electronic health records: A development and validation study," *PLOS Digital Health*, vol. 3, no. 8, p. e0000578, 2024. [Online]. Available: https://doi.org/10.1371/journal.pdig.0000578

[21] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. [Online]. Available: https://doi.org/10.1023/A:1010933404324

[22] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002. [Online]. Available: https://doi.org/10.1613/jair.953

[23] Y. Huang, R. Wu, Y. Huang, Y. Xiang, and W. Zhou, "Investigating the mechanisms of internet gaming disorder and developing intelligent monitoring models using artificial intelligence technologies: Protocol of a prospective cohort," *BMC Public Health*, vol. 24, no. 1, p. 2536, 2024. [Online]. Available: https://doi.org/10.1186/s12889-024-20028-4

[24] sharmajicoder, "Gaming and mental health dataset," https://www.kaggle.com/datasets/sharmajicoder/gaming-and-mental-health, 2025, accessed: 2026-05-24.

[25] S. Skripkauskaite and M. Fazel, "Time spent gaming, device type, addiction scores, and well-being of adolescent English gamers in the 2021 OxWell survey: Latent profile analysis," *JMIR Pediatrics and Parenting*, vol. 5, no. 4, p. e41480, 2022. [Online]. Available: https://doi.org/10.2196/41480